Nick Bukovec

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Education

Stanford University — Stanford, CA
Master of Science, Electrical Engineering, DSP and Optimization (GPA: 3.68) — *Sep. 2024 – March 2026*

California Polytechnic State University, San Luis Obispo — San Luis Obispo, CA
Bachelor of Science, Computer Science (GPA: 3.76) — *Sep. 2020 – June 2024*

Korea University — Seoul, South Korea
Semester Abroad (GPA: 3.91) — *Feb. 2023 – June 2023*

Experience

Software Development Engineer Intern — June 2025 – Sep. 2025
Amazon Web Services — *Seattle, WA*
- Created Model Context Protocol (MCP) server with multimodal support and integrated into LLM agents, enabling developers to query internal data sources through natural language.

Service Desk Student Lead — Aug. 2021 – Sep. 2024
California Polytechnic State University Information Technology Services — *San Luis Obispo, CA*
- Supervised team of forty students that provide technical support for university students, staff and emeritus.
- Facilitated operation of call center handling over one hundred calls per day.
- Resolved hundreds of IT issues (network, security, etc.) through email, phone, and in-person.

Software Engineering and Data Science Intern — June 2022 – Aug. 2022
NASA Jet Propulsion Laboratory — *La Cañada Flintridge, CA*
- Spearheaded creation of extensible web configuration tool with React and Next.js for deploying ground data system infrastructure and applications into AWS VPC.
- Validated geospatial datasets and tested submission and evaluation mechanisms for open data challenges.

Projects

Codec-based Frame Sampling | *Python, FFmpeg, H.264, Gemini* — Sep. 2025 – Dec. 2025
- Exploited and tuned H.264 encoder's scene-change detection as a zero-cost keyframe selector for multimodal LLM inputs, eliminating the need for a separate frame scoring model.
- Benchmarked against uniform sampling on MVBench using Gemini; achieved comparable accuracy with fewer input tokens.

FakeSound Mamba | *Python, Pytorch, Lightning, Amazon Sagemaker, Docker* — Sep. 2024 – Dec. 2024
- Extended Vision Mamba architecture in Pytorch and Lightning to classify and segment general audio clips with inpainted deepfake segments using multi-task learning.
- Constructed custom Sagemaker containers to train models using DDP on EC2 Spot instances.
- When tested using deepfake attacks unseen in the training data, model showed higher clip-level classification accuracy and higher segment-level F1 score than CNN baseline and subjective human evaluation.

srsadmm: Serverless ADMM Solver | *Rust, Tokio, Nalgebra, AWS Lambda* — Apr. 2025 – June 2025
- Created distributed LASSO regression solver applying alternating direction method of multipliers (ADMM) to problems with hundreds of thousands of data points using hybrid computing model.
- Performed expensive matrix multiplications and factorizations in cloud before aggregating variables locally.
- After initial factorization and data upload, fit model to 16GB of data in under 3 minutes using 40 parallel workers.

Technical Skills

Completed Coursework: Machine Learning, Deep Learning, Knowledge Discovery from Data, Convex Optimization II, Software Engineering II, Introduction to Databases, Data Compression, Audio Signal Processing, Perceptual Audio Coding.
Languages: Python, Rust, C, SQL (Postgres, MySQL, DuckDB), Java, HTML/CSS, JavaScript, TypeScript, Node.js.
Frameworks: Pytorch, Lightning, Scikit-learn, FastAPI, React, Next.js, Astro, Svelte.
Libraries: NumPy, Matplotlib, Pandas, TensorBoard, wandb, librosa, SciPy, dspy, tokio, nalgebra, FFmpeg, zstd.
Development/Deployment Tools: Git, Docker, Bash, Github Actions, AWS (S3, Lambda, Sagemaker), Modal.